The Formation of Black Holes and Singularities in Spherically Symmetric Gravitational Collapse

DEMETRIOS CHRISTODOULOU
Courant Institute

1. Introduction

In this paper we study the Einstein equations

$$R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R = 8\pi T_{\mu\nu} \tag{1.1a}$$

where the energy-momentum-stress tensor $T_{\mu\nu}$ is that of a scalar field ϕ:

$$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - \tfrac{1}{2}g_{\mu\nu}\partial^\alpha\phi\partial_\alpha\phi. \tag{1.1b}$$

Equations (1.1a), (1.1b) are equivalent to:

$$R_{\mu\nu} = 8\pi\partial_\mu\phi\partial_\nu\phi. \tag{1.1c}$$

The twice contracted Bianchi identities,

$$\nabla^\mu R_{\mu\nu} - \tfrac{1}{2}\partial_\nu R = 0,$$

imply the conservation law:

$$\nabla^\mu T_{\mu\nu} = 0.$$

Since here,

$$\nabla^\mu T_{\mu\nu} = \partial_\nu\phi\Delta\phi,$$

the conservation law is equivalent to the wave equation for ϕ:

$$\Delta\phi = 0. \tag{1.1d}$$

In this paper we confine ourselves to the spherically symmetric case. However, in the first two sections we show how the spherically symmetric case fits into the general case. The problem has been investigated in [1]–[4]. In these papers attention was confined to the domain of outer communications $\mathcal{U}$ (see Section 4). In [4] it was shown that when the final Bondi mass M_1 is positive, as the retarded time measured by faraway observers tends to infinity, a black hole forms of mass M_1 surrounded by vacuum. However, except for the proof in [1] that $M_1 = 0$ for initial data in a neighborhood of trivial data, the dependence of M_1 on the initial data

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was not investigated. In the present paper we show that if the initial future cone has an annular region such that the ratio of the mass contained in this region to the largest radius is large in comparison to the ratio of the radii minus 1, then a trapped sphere forms in the future, the mass of which is bounded from below by a positive number depending only on the two initial radii (see Section 5). Thus, under the above hypothesis on the initial data, M_1 has a positive lower bound. The hypothesis is satisfied even for initial conditions which are as far from the trapped sphere condition as we wish. However in these situations the time required to form a trapped sphere is correspondingly longer.

In the present paper we go beyond the event horizon $\mathscr{H}$ to study the structure of the apparent horizon $\mathcal{A}$ and the singular boundary $\mathcal{B}$. We thus obtain a complete picture of the solutions. Finally, we obtain a lower bound for the rate of blow up of the curvature at $\mathcal{B}$.

2. Spherical Symmetry

Let Γ be a timelike geodesic with unit future directed tangent vectorfield T:

$$\nabla_T T = 0.$$

Let o be an origin on Γ. We define an optical function u by the following two conditions:

1. $u|_\Gamma$ = arc length on Γ measured from o,
2. The level sets $C^+(u)$ of u are future null geodesic cones with vertices on Γ.

We define the future-directed null geodesic vectorfield

$$l^\mu = -g^{\mu\nu}\partial_\nu u.$$

We have:

$$\nabla_l l = 0,$$

$$-g(l, T)|_\Gamma = Tu|_\Gamma = 1.$$

We define the affine parameter function s by:

$$ls = 1, \qquad s|_\Gamma = 0.$$

The level sets $S_{u,s}$ of s on $C^+(u)$ are 2-spheres. We define the conjugate null vectorfield $\underline{l}$ as follows. At each point q on a given sphere $S_{u,s}$, $\underline{l}$ is the unique future-directed null vector in the orthogonal complement of $T_q S_{u,s}$ such that:

$$g(l, \underline{l}) = -2.$$

We can define an action of the rotation group $SO(3)$ on the spacetime such that Γ is the fixed point set of the action. At each point $p \in \Gamma$, vertex of a future null geodesic cone $C^+(u)$, let Σ_p be the 3-dimensional subspace of the tangent space at p which is orthogonal to T. Then $SO(3)$ on Σ_p is realized as the group of linear transformations on Σ_p leaving the induced inner product invariant. Let $O \in SO(3)$ take the unit vector $N \in \Sigma_p$ into the unit vector $ON \in SO\Sigma_p$. Then for each $s \geqq 0$, O takes the point $\exp\{s(T+N)\} \in S_{u,s}$ into the point $\exp\{s(T+ON)\} \in S_{u,s}$. Thus the cones $C^+(u)$ are invariant by the group and the 2-spheres $S_{u,s}$ are the group orbits. Let Ω be a fundamental vectorfield (generator) of the group action. Then by the above definition Ω is tangential to the 2-spheres $S_{u,s}$ and:

$$[\Omega, l] = 0. \tag{2.1a}$$

The spherically (or rotationally) symmetric case is the case where $SO(3)$ acts as an isometry group on the spacetime. Each fundamental vectorfield Ω is then a Killing vectorfield,

$$\mathcal{L}_\Omega g = 0,$$

and the $S_{u,s}$ are metric 2-spheres of Gauss curvature

$$K = r^{-2},$$

where r is the area radius

$$r(u, s) = \sqrt{\frac{\text{Area}(S_{u,s})}{4\pi}}.$$

In the spherically symmetric case we have:

$$[\Omega, \underline{l}] = 0. \tag{2.1b}$$

For,

$$g([\Omega, \underline{l}], l) = \Omega(g(\underline{l}, l)) - g(\underline{l}, [\Omega, l]) = 0,$$

$$g([\Omega, \underline{l}], \underline{l}) = \tfrac{1}{2}\Omega(g(\underline{l}, \underline{l})) = 0,$$

while if X is a vectorfield tangent to $S_{u,s}$,

$$g([\Omega, \underline{l}], X) = \Omega(g(\underline{l}, X)) - g(\underline{l}, [\Omega, X]) = 0,$$

since $[\Omega, X]$ is also tangent to $S_{u,s}$.

We define the null second fundamental form χ and the conjugate null second fundamental form $\underline{\chi}$ of $S_{u,s}$ to be the 2-covariant tensorfields on $S_{u,s}$ given by:

$$\chi(X, Y) = g(\nabla_X l, Y)$$

$$\underline{\chi}(X, Y) = g(\nabla_X \underline{l}, Y)$$

for arbitrary vectorfields X, Y tangent to $S_{u,s}$. Since,

$$\chi(X, Y) - \chi(Y, X) = -g(l, \nabla_X Y - \nabla_Y X) = 0$$

$$\underline{\chi}(X, Y) - \underline{\chi}(Y, X) = -g(\underline{l}, \nabla_X Y - \nabla_Y X) = 0,$$

as $\nabla_X Y - \nabla_Y X = [X, Y]$ is tangent to $S_{u,s}$, the second fundamental forms are symmetric. Let γ be the induced metric on $S_{u,s}$. Then,

$$(\mathcal{L}_l \gamma)(X, Y) = l(g(X, Y)) - g([l, X], Y) - g(X, [l, Y])$$

$$= g(\nabla_X l, Y) + g(X, \nabla_Y l)$$

$$(\mathcal{L}_{\underline{l}} \gamma)(X, Y) = \underline{l}(g(X, Y)) - g([\underline{l}, X], Y) - g(X, [\underline{l}, Y])$$

$$= g(\nabla_X \underline{l}, Y) + g(X, \nabla_Y \underline{l}).$$

That is:

$$(\mathcal{L}_l \gamma)(X, Y) = 2\chi(X, Y)$$

$$(\mathcal{L}_{\underline{l}} \gamma)(X, Y) = 2\underline{\chi}(X, Y).$$

It follows from equations (2.1a), (2.1b) that in the spherically symmetric case χ and $\underline{\chi}$ are rotationally invariant. For, let Ω be a fundamental vectorfield. Then for arbitrary vectorfields X, Y tangent to $S_{u,s}$ we have:

$$(\mathcal{L}_\Omega \chi)(X, Y) = \Omega(\chi(X, Y)) - \chi([\Omega, X], Y) - \chi(X, [\Omega, Y])$$

$$= g([\Omega, \nabla_X l], Y) - \chi([\Omega, X], Y).$$

Now for a Killing vectorfield Ω and any two vectorfields U, V we have:

$$[\Omega, \nabla_U V] - \nabla_U([\Omega, V]) = \nabla_{[\Omega, U]} V.$$

Hence, by (2.1a),

$$[\Omega, \nabla_X l] = \nabla_{[\Omega, X]} l$$

and

$$g([\Omega, \nabla_X l], Y) = g(\nabla_{[\Omega,X]} l, Y) = \chi([\Omega, X], Y),$$

as $[\Omega, X]$ is tangent to $S_{u,s}$. Therefore:

$$\mathcal{L}_\Omega \chi = 0.$$

Similarly we obtain:

$$\mathcal{L}_\Omega \underline{\chi} = 0.$$

There are no nontrivial symmetric tracefree 2-covariant tensorfields on a metric 2-sphere which are invariant under the rotation group; therefore, in the spherically symmetric case:

(2.2a) $$\chi = \tfrac{1}{2} \operatorname{tr} \chi \gamma,$$

(2.2b) $$\underline{\chi} = \tfrac{1}{2} \operatorname{tr} \underline{\chi} \gamma.$$

Here $\operatorname{tr} \chi$ and $\operatorname{tr} \underline{\chi}$ are constant on each $S_{u,s}$.

We define the torsion 1-form of $S_{u,s}$ by

$$\zeta(X) = \tfrac{1}{2} g(\nabla_X l, \underline{l})$$

for an arbitrary vectorfield X tangent to $S_{u,s}$. In the spherically symmetric case equations (2.1a), (2.1b) imply that ζ is rotationally invariant. For, given any vectorfield X tangent to $S_{u,s}$ we have:

$$\begin{aligned}(\mathcal{L}_\Omega \zeta)(X) &= \Omega(\zeta(X)) - \zeta([\Omega, X]) \\ &= \tfrac{1}{2} g([\Omega, \nabla_X l], \underline{l}) - \zeta([\Omega, X]) \\ &= \tfrac{1}{2} g(\nabla_{[\Omega,X]} l, \underline{l}) - \zeta([\Omega, X]) = 0.\end{aligned}$$

There are no nontrivial 1-forms on a metric 2-sphere which are invariant under the rotation group; therefore, in the spherically symmetric case,

$$\zeta = 0.$$

We shall now show that, in general,

(2.3) $$\nabla_l \underline{l} = -2\zeta.$$

[Here by ζ we mean the vectorfield corresponding to the 1-form ζ.] We have:

$$g(\underline{l}, \nabla_l \underline{l}) = \tfrac{1}{2} l(g(\underline{l}, \underline{l})) = 0,$$

$$g(l, \nabla_l \underline{l}) = -g(\underline{l}, \nabla_l l) = 0.$$

Hence $\nabla_l \underline{l}$ is tangent to the 2-spheres $S_{u,s}$. Now let X be an arbitrary vectorfield tangent to $S_{u,s}$ for some s. We extend it to $C^+(u)$ so that it is tangent to $S_{u,s}$ for all s by

$$[l, X] = 0.$$

Then

$$\zeta(X) = \tfrac{1}{2} g(\nabla_X l, \underline{l}) = \tfrac{1}{2} g(\nabla_l X, \underline{l}) = -\tfrac{1}{2} g(X, \nabla_l \underline{l})$$

which implies (2.3). We define:

$$\underline{\omega} = \tfrac{1}{4} g(\nabla_{\underline{l}} l, \underline{l}).$$

We then have,

$$[\underline{l}, l] = 4\zeta - 2\underline{\omega} l. \tag{2.4}$$

Consequently, the condition $\zeta = 0$, which holds in the spherically symmetric case, implies that the distribution of planes spanned by l and $\underline{l}$ is integrable.

The differential of the affine function s is in general given by

$$\partial_\mu s = \tfrac{1}{2}(\underline{a} l_\mu - \underline{l}_\mu)$$

where

$$\underline{a} = g^{\mu\nu} \partial_\mu s \partial_\nu s.$$

In the spherically symmetric case the function $\underline{a}$ is constant on each 2-sphere $S_{u,s}$. In general,

$$\tfrac{1}{4} g(\nabla_{\underline{l}} \underline{l}, l) = -\underline{\omega},$$

and we define the $S_{u,s}$-tangent 1-form $\underline{\xi}$ by

$$\underline{\xi}(X) = \tfrac{1}{2} g(\nabla_{\underline{l}} \underline{l}, X)$$

for any vectorfield X tangent to $S_{u,s}$. We can relate $\underline{\omega}$ and $\underline{\xi}$ to the derivatives of the function $\underline{a}$. From now on we denote l and $\underline{l}$ when considered as differential operators on functions, as D and $\underline{D}$ respectively. Since

$$\underline{a} = -\underline{D}s,$$

we have:

$$D\underline{a} = -\nabla_l \underline{l}^\mu \partial_\mu s - \underline{l}^\mu l^\nu \nabla_\mu \nabla_\nu s.$$

By (2.3) the first term vanishes. Hence,

$$D\underline{a} = \underline{l}^\mu \nabla_\mu l^\nu \nabla_\nu s,$$

that is:

$$D\underline{a} = -2\underline{\omega}. \tag{2.5a}$$

On the other hand, for any vectorfield X tangent to $S_{u,s}$ we have:

$$\begin{aligned}\nabla_X \underline{a} &= -\nabla_X \underline{l}^\mu \partial_\mu s - \underline{l}^\mu X^\nu \nabla_\mu \nabla_\nu s \\ &= -\tfrac{1}{2}\underline{a} g(\nabla_X \underline{l}, l) + \tfrac{1}{2}\underline{a} g(\nabla_l X, l) - \tfrac{1}{2} g(\nabla_l X, \underline{l}).\end{aligned}$$

Hence:

$$\underline{\xi}(X) = \nabla_X \underline{a}. \tag{2.5b}$$

Therefore in the spherically symmetric case it holds:

$$\underline{\xi} = 0.$$

Consequently, in this case we have

$$d\underline{l} = \underline{\omega}\underline{l} \wedge l$$

[Here $\underline{l}$ and l stand for the corresponding 1-forms]. It follows that in the spherically symmetric case the vectorfield $\underline{l}$ generates incoming null hypersurfaces $C^-(v)$, the integral curves of $\underline{l}$ being the null geodesic generators of $C^-(v)$.

The above imply that the metric of a spherically symmetric spacetime can be expressed in the form

$$g = -2\, du\, ds - \underline{a}\, du^2 + r^2 \mathring{\gamma}$$

where $\mathring{\gamma}$ is the metric of the standard (unit) 2-sphere. The vectorfields l and $\underline{l}$ are given by:

$$l = \frac{\partial}{\partial s},$$

$$\underline{l} = 2\frac{\partial}{\partial u} - \underline{a}\frac{\partial}{\partial s}.$$

We have:

$$D(\text{Area}(S)) = \int_S \text{tr}\,\chi\, d\mu_\gamma, \tag{2.6a}$$

$$\underline{D}(\text{Area}(S)) = \int_S \text{tr}\,\underline{\chi}\, d\mu_\gamma. \tag{2.6b}$$

We find convenient to define the dimensionless quantities

$$h = \frac{r}{2}\text{tr}\,\chi,$$

$$\underline{h} = \frac{r}{2}\text{tr}\,\underline{\chi},$$

in terms of which we can express equations (2.6a), (1.5b) in the form:

$$Dr = h, \tag{2.6e}$$

$$\underline{D}r = \underline{h}. \tag{2.6d}$$

The optical geometry in the general case is discussed in [5].

3. Derivation of the Equations

From now on our attention is confined to the spherically symmetric case. At each $p \in \Gamma$, let $(e_A : A = 1, 2)$ be a local orthonormal frame for the unit 2-sphere in Σ_p. We parallel propagate this frame along the generators of the future null geodesic cone with vertex at p:

$$\nabla_l e_A = 0.$$

Then for each u and s, $(e_A : A = 1, 2)$ is an orthonormal frame for the 2-sphere $S_{u,s}$ and $(e_A : A = 1, 2; e_3 = \underline{l}, e_4 = l)$ is a null frame for the spacetime. In the

following the components of tensorfields in spacetime are relative to such a null frame.

We have:

$$D\chi_{AB} = e_A^\mu e_B^\nu \nabla_l \nabla_\mu l_\nu$$

$$= e_A^\mu e_B^\nu l^\alpha (\nabla_\mu \nabla_\alpha l_\nu - R_{\nu\beta\mu\alpha} l^\beta)$$

$$= e_A^\mu e_B^\nu (-\nabla_\mu l^\alpha \nabla_\alpha l_\nu - R_{\mu\alpha\nu\beta} l^\alpha l^\beta),$$

that is:

$$D\chi_{AB} = -\chi_{AC}\chi_{CB} - R_{A4B4}. \tag{3.1a}$$

By (2.2a) the trace-free part of this equation expresses the vanishing of the trace-free part of R_{A4B4}. Hence

$$R_{A4B4} = \tfrac{1}{2} R_{44} \delta_{AB} = 4\pi\psi^2 \delta_{AB} \tag{3.1b}$$

where

$$\psi = D\phi.$$

The trace of equation (3.1a) is the equation

$$D \operatorname{tr} \chi = -\tfrac{1}{2}(\operatorname{tr} \chi)^2 - R_{44}. \tag{3.1c}$$

That is:

$$Dh = -4\pi r \psi^2. \tag{3.1d}$$

Given ψ on $C^+(u)$, this equation is a linear second-order ordinary differential equation for r on $C^+(u)$:

$$D^2 r + 4\pi r \psi^2 = 0. \tag{3.1e}$$

We have:

$$\underline{D}\underline{\chi}_{AB} = (e_B^\nu \nabla_{\underline{l}} e_A^\mu + e_A^\mu \nabla_{\underline{l}} e_B^\nu) \nabla_\mu \underline{l}_\nu + e_A^\mu e_B^\nu \nabla_{\underline{l}} \nabla_\mu \underline{l}_\nu.$$

Now,

$$(e_B^\nu \nabla_{\underline{l}} e_A^\mu + e_A^\mu \nabla_{\underline{l}} e_B^\nu) \nabla_\mu \underline{l}_\nu = \underline{\chi}_{CB} g(e_C, \nabla_{\underline{l}} e_A) + \underline{\chi}_{CA} g(e_C, \nabla_{\underline{l}} e_B)$$

$$= \tfrac{1}{2} \operatorname{tr} \underline{\chi} \underline{D}(g(e_A, e_B)) = 0.$$

Hence:

$$\begin{aligned}\underline{D}\underline{\chi}_{AB} &= e^{\mu}_{A}e^{\nu}_{B}\nabla_{\underline{l}}\nabla_{\mu}\underline{l}_{\nu}\\ &= e^{\mu}_{A}e^{\nu}_{B}\underline{l}^{\alpha}(\nabla_{\mu}\nabla_{\alpha}\underline{l}_{\nu} - R_{\nu\beta\mu\alpha}\underline{l}^{\beta})\\ &= e^{\mu}_{A}e^{\nu}_{B}(\nabla_{\mu}(\nabla_{\underline{l}}\underline{l}_{\nu}) - \nabla_{\mu}\underline{l}^{\alpha}\nabla_{\alpha}\underline{l}_{\nu} - R_{\mu\alpha\nu\beta}\underline{l}^{\alpha}\underline{l}^{\beta}).\end{aligned}$$

Since

$$\nabla_{\underline{l}}\underline{l} = 2\underline{\omega}\underline{l},$$

we have,

$$e^{\mu}_{A}e^{\nu}_{B}\nabla_{\mu}(\nabla_{\underline{l}}\underline{l}_{\nu}) = 2\underline{\omega}\underline{\chi}_{AB}.$$

Therefore we obtain:

$$\underline{D}\underline{\chi}_{AB} = -\underline{\chi}_{AC}\underline{\chi}_{CB} + 2\underline{\omega}\underline{\chi}_{AB} - R_{A3B3}. \tag{3.2a}$$

By (2.2b) the trace-free part of this equation expresses the vanishing of the trace-free part of R_{A3B3}. Hence

$$R_{A3B3} = \tfrac{1}{2}R_{33}\delta_{AB} = 4\pi\underline{\psi}^{2}\delta_{AB} \tag{3.2b}$$

where

$$\underline{\psi} = \underline{D}\phi.$$

The trace of equation (3.2a) is the equation

$$\underline{D}\ \mathrm{tr}\ \underline{\chi} - 2\underline{\omega}\ \mathrm{tr}\ \underline{\chi} = -\tfrac{1}{2}(\mathrm{tr}\ \underline{\chi})^{2} - R_{33}. \tag{3.2c}$$

That is,

$$\underline{D}\underline{h} - 2\underline{\omega}\underline{h} = -4\pi r\underline{\psi}^{2}. \tag{3.2d}$$

In view of the fact that

$$\nabla_{l}\underline{l} = 0$$

we have

$$D\underline{\chi}_{AB} = e^{\mu}_{A} e^{\nu}_{B} \nabla_{l} \nabla_{\mu} \underline{l}_{\nu}$$

$$= e^{\mu}_{A} e^{\nu}_{B} l^{\alpha} (\nabla_{\mu} \nabla_{\alpha} \underline{l}_{\nu} - R_{\nu\beta\mu\alpha} \underline{l}^{\beta})$$

$$= e^{\mu}_{A} e^{\nu}_{B} (-\nabla_{\mu} l^{\alpha} \nabla_{\alpha} \underline{l}_{\nu} - R_{\mu\alpha\nu\beta} l^{\alpha} \underline{l}^{\beta}),$$

that is,

$$D\underline{\chi}_{AB} = -\chi_{AC} \underline{\chi}_{CB} - R_{A4B3}. \tag{3.3a}$$

Consider

$$\tfrac{1}{2} R_{AB34} = \sigma \epsilon_{AB}$$

where ϵ is the area 2-form of $S_{u,s}$. The antisymmetric part of R_{A3B4} is given by

$$\tfrac{1}{2}(R_{A3B4} - R_{B3A4}) = \sigma \epsilon_{AB}.$$

Thus, by (2.2a), (2.2b), the antisymmetric part of equation (3.3a) is equivalent to

$$\sigma = 0.$$

The Gauss equation of the imbedding of $S_{u,s}$ in spacetime in general reads

$$K + \tfrac{1}{4} \operatorname{tr} \chi \operatorname{tr} \underline{\chi} - \tfrac{1}{2} \hat{\chi} \cdot \hat{\underline{\chi}} = -\rho$$

where

$$R_{ABCD} = -\rho \epsilon_{AB} \epsilon_{CD}$$

and $\hat{\chi}$, $\hat{\underline{\chi}}$ are the trace-free parts of χ, $\underline{\chi}$ respectively. In the spherically symmetric case the Gauss equation reads:

$$r^{-2}(1 + h\underline{h}) = -\rho. \tag{3.3b}$$

Also,

$$R^{A}_{BAD} + R^{3}_{B3D} + R^{4}_{B4D} = R_{BD} = 0,$$

which expresses the symmetric part of R_{A3B4} as

$$\tfrac{1}{2}(R_{A3B4} + R_{B3A4}) = -\rho \delta_{AB}.$$

Therefore the symmetric part of equation (3.3a) is equivalent to

$$D \operatorname{tr} \underline{\chi} = -\tfrac{1}{2} \operatorname{tr} \chi \operatorname{tr} \underline{\chi} + 2\rho. \tag{3.3c}$$

Substituting from (3.3b) we then obtain

$$D\underline{h} + r^{-1}h\underline{h} = -r^{-1}. \tag{3.3d}$$

An immediate consequence of this equation and the commutation relation

$$[\underline{D}, D] = -2\underline{\omega}D \tag{3.4}$$

is the equation

$$\underline{D}h + 2\underline{\omega}h = D\underline{h}. \tag{3.5}$$

We now derive an equation for $\underline{\omega}$ on each $C^+(u)$. We have

$$\begin{aligned} D\underline{\omega} &= \tfrac{1}{4}\underline{l}^\mu \underline{l}^\nu \nabla_l \nabla_\mu l_\nu \\ &= \tfrac{1}{4}\underline{l}^\mu \underline{l}^\nu l^\alpha (\nabla_\mu \nabla_\alpha l_\nu - R_{\nu\beta\mu\alpha} l^\beta). \end{aligned}$$

Since

$$\underline{l}^\mu \underline{l}^\nu \nabla_\mu l^\alpha \nabla_\alpha l_\nu = 0,$$

this reduces to

$$D\underline{\omega} = -\tfrac{1}{4}R_{3434}. \tag{3.6a}$$

On the other hand,

$$R^A_{3A4} + R^3_{334} = R_{34} = 8\pi\psi\underline{\psi}.$$

Hence

$$\tfrac{1}{4}R_{3434} = 4\pi\psi\underline{\psi} + \rho. \tag{3.6b}$$

Substituting (3.3b) in (3.6b) and the result in equation (3.6a) we conclude:

$$D\underline{\omega} = r^{-2}(1 + h\underline{h}) - 4\pi\psi\underline{\psi}. \tag{3.6c}$$

The null structure equations in the general case are discussed in [5].

The components $\tfrac{1}{2}R_{A334} = \underline{\beta}_A$, $\tfrac{1}{2}R_{A434} = \beta_A$ of the curvature tensor define 1-forms $\underline{\beta}$, β on the 2-spheres $S_{u,s}$. In the spherically symmetric case these 1-forms

are rotationally invariant; since there are no nontrivial 1-forms on a metric 2-sphere which are invariant under the rotation group, we have

$$\underline{\beta} = \beta = 0.$$

The remaining components of the curvature tensor R_{ABC3}, R_{ABC4} also vanish as we can express

$$R_{ABC3} = \epsilon_{AB}\epsilon_{CD}(\underline{\beta}_D + R_{D3}),$$

$$R_{ABC4} = \epsilon_{AB}\epsilon_{CD}(-\beta_D + R_{D4}).$$

It follows from the above that

$$\tfrac{1}{4}R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} = \Sigma_0 + \Sigma_2 + \Sigma_4 \tag{3.7}$$

where

$$\Sigma_0 = \tfrac{1}{4}R_{ABCD}R_{ABCD} = \rho^2,$$

$$\Sigma_2 = \tfrac{1}{2}(R_{A3B3}R_{A4B4} + R_{A3B4}R_{A4B3}) = (4\pi\psi\underline{\psi})^2 + \rho^2,$$

$$\Sigma_4 = (\tfrac{1}{4}R_{3434})^2 = (4\pi\psi\underline{\psi} + \rho)^2.$$

We shall now express, in general, the wave equation

$$\Delta\phi = g^{\mu\nu}\nabla_\mu\nabla_\nu\phi = 0$$

in terms of the foliation of spacetime given by the 2-spheres $S_{u,s}$. Let $\prod$ be the projection to $S_{u,s}$:

$$\prod{}^{\mu}_{\nu} = \delta^{\mu}_{\nu} + \tfrac{1}{2}(\underline{l}^{\mu}l_\nu + l^{\mu}\underline{l}_\nu).$$

Then the wave equation can be expressed in the form

$$\prod{}^{\mu\nu}\nabla_\mu\nabla_\nu\phi = \tfrac{1}{2}(\nabla_{\underline{l}}\psi + \nabla_l\underline{\psi}) - \tfrac{1}{2}(\nabla_{\underline{l}}l^\nu + \nabla_l\underline{l}^\nu)\nabla_\nu\phi.$$

Let us denote by $\not\nabla$ the operator of covariant differentiation intrinsic to the 2-spheres $S_{u,s}$. We have:

$$\nabla_\nu\phi = \not\nabla_\nu\phi - \tfrac{1}{2}(\underline{l}_\nu\psi + l_\nu\underline{\psi}).$$

Let $\not\Delta$ be the Laplacian intrinsic to $S_{u,s}$:

$$\Delta\phi = \prod^{\mu\nu} \nabla_\nu \nabla_\nu \phi.$$

Substituting we find:

$$\prod^{\mu\nu} \nabla_\mu \nabla_\nu \phi = \Delta\phi - \tfrac{1}{2}(\mathrm{tr}\, \underline{\chi}\psi + \mathrm{tr}\, \chi\underline{\psi}).$$

Since also

$$\tfrac{1}{2}(\nabla_l l + \nabla_{\underline{l}} \underline{l}) = -\underline{\omega} l$$

the wave equation, in general, reads

$$\not\Delta\phi - \tfrac{1}{2}(\mathrm{tr}\, \underline{\chi}\psi + \mathrm{tr}\, \chi\underline{\psi}) = \tfrac{1}{2}(\underline{D}\psi + D\underline{\psi}) + \underline{\omega}\psi.$$

In the spherically symmetric case this becomes

$$\underline{D}\psi + D\underline{\psi} + 2\underline{\omega}\psi + 2r^{-1}(\underline{h}\psi + h\underline{\psi}) = 0.$$

In view of the fact that by the commutation relation (3.4)

$$\underline{D}\psi + 2\underline{\omega}\psi = D\underline{\psi} \tag{3.8a}$$

the wave equation in the spherically symmetric case reduces to

$$D\underline{\psi} + r^{-1}(h\underline{\psi} + \underline{h}\psi) = 0. \tag{3.8b}$$

4. The Mass

The system of equations which we have derived is:

$$Dr = h, \tag{4.1a}$$

$$\underline{D}r = \underline{h}, \tag{4.1b}$$

$$Dh = -4\pi r\psi^2, \tag{4.1c}$$

$$D\underline{h} + r^{-1}h\underline{h} = -r^{-1}, \tag{4.1d}$$

$$\underline{D}h + 2\underline{\omega}h + r^{-1}\underline{h}h = -r^{-1}, \tag{4.1e}$$

$$\underline{D}\underline{h} - 2\underline{\omega}\underline{h} = -4\pi r\underline{\psi}^2, \tag{4.1f}$$

(4.1g) $$D\underline{\omega} = r^{-2}(1 + h\underline{h}) - 4\pi\psi\underline{\psi},$$

(4.1h) $$D\underline{a} = -2\underline{\omega},$$

(4.1i) $$D\underline{\psi} + r^{-1}h\underline{\psi} = -r^{-1}\underline{h}\psi,$$

(4.1j) $$\underline{D}\psi + 2\underline{\omega}\psi + r^{-1}\underline{h}\psi = -r^{-1}h\underline{\psi}.$$

Equations (4.1a), (4.1c), (4.1g), (4.1h) need to be supplemented with boundary conditions on Γ. The boundary condition for equation (4.1a) is

(4.2a) $$r|_\Gamma = 0$$

which, in view of the fact that

$$\tfrac{1}{2}(l + \underline{l})|_\Gamma = T,$$

implies

(4.2b) $$\nabla_T r = \tfrac{1}{2}(h + \underline{h})|_\Gamma = 0.$$

Equation (4.1d) can be written in the form

$$D(r\underline{h}) = -1.$$

Integrating on each $C^+(u)$ from the vertex, taking into account (4.2a), we obtain

(4.2c) $$\underline{h} = -\frac{s}{r}.$$

Hence, again by (4.2a),

$$\underline{h}|_\Gamma = -\frac{1}{h|_\Gamma}$$

which together with (4.2b) yields the boundary condition for equation (4.1c):

(4.2d) $$h|_\Gamma = 1.$$

As Γ is the set where $s = 0$ we have

$$Ts = 0.$$

This is equivalent to the following boundary condition for equation (4.1h):

(4.2e) $$\underline{a}|_\Gamma = 1.$$

To obtain the boundary condition for $\underline{\omega}$ we consider the fact that

$$\tfrac{1}{4}\nabla_{l+\underline{l}}(l+\underline{l}) = -\tfrac{1}{2}(l-\underline{l})\underline{\omega},$$

while

$$\tfrac{1}{4}\nabla_{l+\underline{l}}(l+\underline{l})|_{\Gamma} = \nabla_T T = 0.$$

Thus,

$$\underline{\omega}|_{\Gamma} = 0. \tag{4.2f}$$

Finally, equation (4.1i) can be written in the form:

$$D(r\underline{\psi}) = -\underline{h}\psi.$$

In view of (4.2a), integrating on each $C^+(u)$ from the vertex yields

$$\underline{\psi} = -\frac{1}{r}\int_0^s \underline{h}\psi \, ds. \tag{4.2g}$$

Given ψ on a future null geodesic cone $C^+(u)$, equations (4.1a), (4.1c), together with the boundary conditions (4.2a), (4.2d), determine r and h on $C^+(u)$. Equation (4.1d) then determines $\underline{h}$ (4.2c) and equation (4.1i) determines $\underline{\psi}$ (4.2g). Then $\underline{\omega}$ is determined on $C^+(u)$ from equation (4.1g) together with the boundary condition (4.2f) after which equation (4.1h) and the boundary condition (4.2e) determine $\underline{a}$. Thus the operator $\underline{D}$ is determined on $C^+(u)$. Finally equation (4.1j) evolves ψ to the future of $C^+(u)$. The initial data are to give ψ as a function of s on the initial future cone C_0^+.

We shall now show that equations (4.1b), (4.1e), and (4.1f) follow from the other equations and the boundary conditions. First, equation (4.1e) is a consequence of equations (4.1b), (4.1d), and the commutation relation (3.4). Let

$$f = \underline{D}r - \underline{h}.$$

Then, by the commutation relation (3.4),

$$Df = \underline{D}h + 2\underline{\omega}h - D\underline{h},$$

$$D^2 f = \underline{D}Dh + 4\underline{\omega}Dh + 2hD\underline{\omega} - D^2\underline{h}.$$

Applying $\underline{D}$ to equation (4.1c) we obtain

$$\underline{D}Dh = -4\pi\psi^2 f - 4\pi\underline{h}\psi^2 - 8\pi r\psi\underline{D}\psi.$$

Hence by equation (4.1j),

$$\underline{D}Dh = -4\pi\psi^2 f + 4\pi\underline{h}\psi^2 + 8\pi h\psi\underline{\psi} + 16\pi r\psi^2\underline{\omega}.$$

Also, applying D to equation (4.1d) and using equation (4.1c) we obtain

$$D^2\underline{h} = r^{-2}h(1 + h\underline{h}) - r^{-1}(\underline{h}Dh + hD\underline{h}),$$

$$= 2r^{-2}h(1 + h\underline{h}) + 4\pi\underline{h}\psi^2.$$

Substituting above and also taking into account equation (4.1g) we find

$$D^2 f + 4\pi\psi^2 f = 0.$$

Now the boundary conditions (4.2a), (4.2d), and (4.2e) imply:

$$f|_\Gamma = Df|_\Gamma = 0.$$

It follows that f vanishes, that is, equation (4.1b) holds. Let

$$g = \underline{D}\underline{h} - 2\underline{\omega}\underline{h}.$$

Then, by the commutation relation (3.4),

$$D(rg) = r\underline{D}D\underline{h} - 2r\underline{h}D\underline{\omega} + h(\underline{D}h - 2\underline{\omega}\underline{h}).$$

Substituting from equations (4.1d), (4.1e), and (4.1g) we obtain

$$D(rg) = 8\pi\underline{h}\psi\underline{\psi}.$$

On the other hand, by equation (4.1i),

$$D(4\pi r^2\underline{\psi}^2) = -8\pi r\underline{h}\psi\underline{\psi}.$$

Hence

$$D(rg + 4\pi r^2\underline{\psi}^2) = 0$$

which, integrated on each $C^+(u)$ from the vertex, taking into account boundary condition (4.2a), gives equation (4.1f).

The Hawking mass $m(S)$ contained within a spacelike 2-sphere S is, in general, defined by

$$m(S) = \left(\frac{r(S)}{2}\right)\left(1 + \frac{1}{16\pi}\int_S \operatorname{tr}\chi \operatorname{tr}\underline{\chi}\, d\mu_\gamma\right)$$

where

$$r(S) = \sqrt{\frac{\text{Area}(S)}{4\pi}}.$$

For the 2-spheres $S_{u,s}$, in the spherically symmetric case, this becomes:

$$m = \frac{r}{2}(1 + h\underline{h}). \tag{4.3}$$

The mass satisfies the following basic equations:

$$Dm = -2\pi r^2 \underline{h}\psi^2, \tag{4.4a}$$

$$\underline{D}m = -2\pi r^2 h\underline{\psi}^2. \tag{4.4b}$$

The first equation follows, in a straightforward manner, from equations (4.1a), (4.1c), (4.1d), while the second follows similarly from equations (4.1b), (4.1e), (4.1f). In the following the spacetime is the future of the initial future null geodesic cone $C_0^+ = C^+(0)$.

LEMMA 1. *The inequality*

$$\underline{h} \leqq -1$$

holds everywhere.

Proof: Equations (4.1a), (4.1c) imply that r is a concave function of s on each $C^+(u)$. The boundary conditions (4.2a), (4.2d) then imply

$$r \leqq s.$$

Hence, by equation (4.2c) the result follows.

PROPOSITION 4.1. *We have*

$$m \geqq 0$$

and

$$m(u_1, s_1) = 0$$

if and only if $\mathcal{D}(S_{u_1,s_1})$, the domain of dependence of S_{u_1,s_1}, is flat and ϕ is constant in $\mathcal{D}(S_{u_1,s_1})$.

Proof: In view of Lemma 1, the first statement follows from equation (4.4a) and the fact that by (4.2a)

$$m|_\Gamma = 0.$$

Now suppose that

$$m(u_1, s_1) = 0. \tag{4.5}$$

Then in the interior $C^+(u_1, s_1)$ of the sphere S_{u_1,s_1} in $C^+(u_1)$ we have

$$\psi = 0.$$

Therefore

$$h(u_1, s_1) = 1.$$

Consider now the incoming null hypersurface C_1^- intersecting $C^+(u_1)$ at S_{u_1,s_1}. Let C_1^- intersect $C^+(u)$ at $S_{u,\lambda(u)}$. Then for some $\epsilon > 0$, $h(u, \lambda(u)) > 0$ for all $u \in [u_1, u_1 + \epsilon]$. Let

$$u_2 = \sup\{u : \forall u' \in [u_1, u]\lambda(u') > 0 \quad \text{and} \quad h(u', \lambda(u')) > 0\}.$$

Then if $\lambda(u_2) > 0$ we must have $h(u_2, \lambda(u_2)) = 0$ while $h(u, \lambda(u)) > 0$ for all $u \in [u_1, u_2)$. The equation (4.4b) along C_1^- together with the first statement of the proposition and (4.5) implies that

$$m(u, \lambda(u)) = 0 : \quad \forall u \in [u_1, u_2].$$

Therefore $\psi = 0$ in $C^+(u, \lambda(u))$ for all $u \in [u_1, u_2]$, which implies

$$h(u, \lambda(u)) = 1 : \quad \forall u \in [u_1, u_2];$$

this shows that in fact $\lambda(u_2) = 0$, that is, C_1^- is a past null geodesic cone with vertex at u_2 on Γ. Furthermore,

$$m(u, \lambda(u)) = 0 : \quad \forall u \in [u_1, u_2]$$

which implies that

$$\psi = \underline{\psi} = 0,$$

$$h = 1, \quad \underline{h} = -1, \quad \underline{\omega} = 0, \quad \underline{a} = 1$$

in

$$\mathcal{D}(S_{u_1,s_1}) = \bigcup_{u\in[u_1,u_2]} C^+(u, \lambda(u)).$$

The above proposition has the following important consequence:

PROPOSITION 4.2. *Consider two outgoing null hypersurfaces C_1^+ and C_2^+, with C_2^+ in the future of C_1^+, and two incoming null hypersurfaces C_1^- and C_2^-, with C_2^- in the exterior of C_1^-. Let $S_{1,1}$, $S_{2,1}$, $S_{1,2}$, $S_{2,2}$ be the spheres*:

$$S_{i,j} = C_i^- \cap C_j^+ : i, j = 1, 2.$$

Then, if on C_2^- $h > 0$, we have

$$\frac{r_{2,2} - r_{1,2}}{r_{2,1} - r_{1,1}} \leqq 1$$

with equality holding if and only if $\mathcal{D}(S_{2,1})$ is flat.

Proof: Let $C_1^+ = C^+(u_1)$, $C_2^+ = C^+(u_2)$. We consider the incoming null hypersurfaces $C^-(v)$ intersecting C_1^+ in spheres $S_{u_1,v}$ contained in the annular region bounded by $S_{1,1}$ and $S_{2,1}$. Here v is the affine parameter value corresponding to $S_{u_1,v}$. We denote by v_1 and v_2 the affine parameter values corresponding to $S_{1,1}$ and $S_{2,1}$ respectively. Let

$$S(u, v) = C^+(u) \cap C^-(v)$$

and let

$$r(u, v) = \text{ radius of } S(u, v).$$

[Note that here r is considered to be a function of (u, v), not of (u, s).] Then we have

$$\frac{\partial r}{\partial u} = \frac{1}{2} \underline{D} r = \frac{1}{2} \underline{h},$$

$$\frac{\partial^2 r}{\partial u \partial v} = \frac{1}{2} D\underline{h} \frac{\partial}{\partial v} s|_{C+(u)},$$

and

$$\frac{\partial}{\partial v} s|_{C+(u)} = \left(\frac{\partial}{\partial s} r|_{C+(u)}\right)^{-1} \frac{\partial r}{\partial v} = h^{-1} \frac{\partial r}{\partial v}.$$

Hence,

$$\frac{\partial^2 r}{\partial u \partial v} = \frac{1}{2} h^{-1} D\underline{h} \frac{\partial r}{\partial v}.$$

In view of definition (4.3), equation (4.1d) takes the form

$$D\underline{h} = -\frac{2m}{r^2}. \tag{4.6}$$

We thus have

$$\frac{\partial^2 r}{\partial u \partial v} = -\frac{1}{h}\frac{m}{r^2}\frac{\partial r}{\partial v}. \tag{4.7}$$

Now, h is decreasing along each $C^+(u)$. Hence our assumption implies that for each $u \in [u_1, u_2]$ $h \geqq 0$ in the interior of $S(u, v_2)$ in $C^+(u)$. Proposition 4.1 together with this fact implies that

$$f(u, v) := \left(\frac{1}{h}\frac{m}{r^2}\right)\Big|_{S(u,v)} \geqq 0$$

for all $(u, v) \in [u_1, u_2] \times [v_1, v_2]$. Thus, integrating equation (4.7) with respect to u on $[u_1, u_2]$,

$$\frac{\partial r}{\partial v}(u_2, v) = \exp\left\{-\int_{u_1}^{u_2} f(u, v)\, du\right\} \frac{\partial r}{\partial v}(u_1, v),$$

we conclude that

$$\frac{\partial r}{\partial v}(u_2, v) \leqq \frac{\partial r}{\partial v}(u_1, v),$$

which upon integrating with respect to v on $[v_1, v_2]$ yields the inequality of the proposition. Furthermore, the case of equality holds if and only if m vanishes in the spacetime region bounded by the four null hypersurfaces. By Proposition 4.1 this is equivalent to $\mathcal{D}(S_{2,1})$ being flat.

In the proof of the collapse theorem we shall use, in addition to the above, the following two lemmas.

LEMMA 2. *Consider, in an outgoing null hypersurface C^+, the annular region bounded by two spheres S_1 and S_2, with S_2 in the exterior of S_1. Let*

$$\eta = \frac{2(m_2 - m_1)}{r_2}.$$

Then if $h_2 > 0$ *we have*

$$\frac{h_1}{h_2} \geqq e^{\eta}.$$

Proof: Let s_1 and s_2 be the affine parameter values on C^+ corresponding to S_1 and S_2 respectively. The assumption $h_2 > 0$ implies $h > 0$ in the interior of S_2 on C^+. We can thus multiply equation (4.1c) by h^{-1} and integrate with respect to s on $[s_1, s_2]$ to obtain

$$\log\left(\frac{h_1}{h_2}\right) = 4\pi \int_{s_1}^{s_2} h^{-1}\psi^2 r \, ds.$$

Using equation (4.4a) we can express this in the form

$$\log\left(\frac{h_1}{h_2}\right) = 2 \int_{s_1}^{s_2} r^{-1}(-h\underline{h})^{-1} Dm \, ds.$$

Now Proposition 4.1 implies

$$-h\underline{h} \leqq 1.$$

It follows that

$$\log\left(\frac{h_1}{h_2}\right) \geqq 2r_2^{-1} \int_{s_1}^{s_2} Dm \, ds = \eta.$$

LEMMA 3. *Consider, in an outgoing null hypersurface* C^+, *the annular region bounded by two spheres* S_1 *and* S_2, *with* S_2 *in the exterior of* S_1. *We define*

$$\eta = \frac{2(m_2 - m_1)}{r_2},$$

$$\theta = r_2\underline{\psi}_2 - r_1\underline{\psi}_1.$$

Then if $h_2 > 0$ *we have*

$$2\pi\left(-\frac{h_2}{\underline{h}_2}\right)\theta^2 \leqq \frac{\eta}{2}\left(\frac{r_2}{r_1} - 1\right).$$

Proof: By equation (4.2g),

$$\theta = -\int_{s_1}^{s_2} \underline{h}\psi \, ds.$$

Hence

$$\theta^2 \leqq \left(\int_{s_1}^{s_2} r^2(-\underline{h})\psi^2\, ds\right)\left(\int_{s_1}^{s_2} r^{-2}(-\underline{h})\, ds\right). \tag{4.8a}$$

By equation (4.4a),

$$\int_{s_1}^{s_2} r^2(-\underline{h})\psi^2\, ds = \frac{1}{2\pi}\int_{s_1}^{s_2} Dm\, ds = \frac{m_2 - m_1}{2\pi}. \tag{4.8b}$$

The assumption $h_2 > 0$ implies $h > 0$ in the interior of S_2 on C^+; also, both h and $\underline{h}$ are decreasing along C^+. Therefore

$$\begin{aligned}\int_{s_1}^{s_2} r^{-2}(-\underline{h})\, ds &= \int_{r_1}^{r_2} r^{-2}(-\underline{h})h^{-1}\, dr \\ &\leqq (-\underline{h}_2)h_2^{-1}\int_{r_1}^{r_2} r^{-2}\, dr = (-\underline{h}_2)h_2^{-1}\left(\frac{1}{r_1} - \frac{1}{r_2}\right).\end{aligned} \tag{4.8c}$$

Substituting (4.8b) and (4.8c) in (4.8a) yields the lemma.

In the following it is assumed that the initial future null geodesic cone C_0^+ intersects future null infinity; that is, the generators of C_0^+ extend to infinity in affine parameter and $r \to \infty$ as $s \to \infty$ on C_0^+. We consider the set of real numbers U defined by

$$U = \{u \geqq 0 : s \quad \text{extends to} \quad \infty \quad \text{on} \quad C^+(u) \quad \text{and} \quad r(u, s) \to \infty \quad \text{as} \quad s \to \infty\}.$$

That is, U is the set of future null geodesic cones, with vertices on Γ, which intersect future null infinity. We remark that $u \in U$ implies that $[0, u] \subset U$. The domain of outer communications $\mathcal{U}$ is defined by

$$\mathcal{U} = \bigcup_{u \in U} C^+(u).$$

In the following we assume that on the initial cone C_0^+ the mass function m has an upper bound. We define the initial Bondi mass M_0 by

$$M_0 = \sup_{s \geqq 0} m(0, s).$$

Then equations (4.4a), (4.4b) imply that for each $u \in U$ we can define the Bondi mass

$$M(u) = \lim_{s \to \infty} m(u, s)$$

and $M(u)$ is a decreasing function of u. We have $M(0) = M_0$. The final Bondi mass M_1 is defined by

$$M_1 = \inf_{u \in U} M(u).$$

If the set U has an upper bound, we define

$$u_{\mathscr{H}} = \sup U.$$

Then, the future null geodesic cone $C^+(u_{\mathscr{H}})$ is the event horizon, which we denote simply by $\mathscr{H}$. In [4] it was shown that if $M_1 > 0$ then U has an upper bound and on $\mathscr{H}$ it holds:

$$\lim_{s \to \infty} r(s) = 2M_1.$$

This implies:

PROPOSITION 4.3. *On* $\mathscr{H}$:

$$\lim_{s \to \infty} m(s) = M_1.$$

Proof: Suppose, on the contrary, that

$$\lim_{s \to \infty} m(s) = L \neq M_1.$$

Then, in view of the fact that by (4.3) and (4.2c),

$$sh = r - 2m,$$

while $r \to 2M_1$, it is necessary that $L < M_1$, for otherwise h would be negative for large enough s, which implies that r becomes zero; a contradiction. Since $sh \to 2(M_1 - L)$ we have $sh \geqq M_1 - L$ for $s \geqq s_1$ (large enough), hence

$$r(s) = r(s_1) + \int_{s_1}^{s} h \, ds \geqq r(s_1) + (M_1 - L)\log\left(\frac{s}{s_1}\right)$$

which contradicts the fact that $r \to 2M_1$ as $s \to \infty$ on $\mathscr{H}$.

For each $u \in U$ and $R \geqq 0$ there is a unique 2-sphere $S_R(u)$ in $C^+(u)$ defined by

$$S_R(u) = S_{u,s} : r(u, s) = R.$$

Let $m_R(u)$ be the mass contained within $S_R(u)$. One of the main results of [4] was that for each $R > 2M_1$, $M(u) - m_R(u)$, the mass remaining in the exterior of $S_R(u)$ in $C^+(u)$ tends to zero as $u \to u_{\mathscr{I}}$.

5. The Collapse Theorem

THEOREM 5.1. *Consider on the initial future null geodesic cone C_0^+ an annular region bounded by two spheres $S_{1,0}$ and $S_{2,0}$ with $S_{2,0}$ in the exterior of $S_{1,0}$. Let*

$$\delta_0 := \frac{r_{2,0}}{r_{1,0}} - 1 \in \left(0, \frac{1}{2}\right).$$

We define:

$$\eta_0 = \frac{2(m_{2,0} - m_{1,0})}{r_{2,0}}.$$

Let E be the following function on the positive real line:

$$E(y) = \frac{y}{(1+y)^2}\left[\log\left(\frac{1}{2y}\right) + 5 - y\right].$$

[*Note*: $E(0) = 0$, $E(\frac{1}{2}) = 1$, *E is increasing and concave on* $(0, \frac{1}{2})$.] *Then if*

$$\eta_0 \geqq E(\delta_0)$$

the following conclusions hold:

(a) *Consider the incoming null hypersurface C_2^- intersecting C_0^+ at $S_{2,0}$. Then there is a future null geodesic cone C_*^+, lying in the future of C_0^+, such that the sphere $S_{2*} = C_2^- \cap C_*^+$ has maximal area in C_*^+ : $h_{2*} = 0$.*

(b) *The radius of S_{2*} is bounded from below by*

$$r_{2*} \geqq \frac{3\delta_0}{(1+\delta_0)} r_{2,0},$$

while the mass m_{2*} contained within S_{2*} is equal to

$$m_{2*} = \frac{r_{2*}}{2}.$$

(c) *The incoming null hypersurface C_1^- through $S_{1,0}$ intersects C_*^+ at a sphere S_{1*} of radius*

$$r_{1*} \geqq 2\delta_0 r_{1,0}.$$

(d) *The temporal distance between the vertices of C_0^+ and C_*^+, which is equal to the length of the corresponding segment of Γ, is bounded from above by*

$$u_* \leqq 2(1 - 2\delta_0) r_{1,0}.$$

(e) *There is an event horizon $\mathscr{H}$ not later than C_*^+. The final Bondi mass*

$$M_1 \geqq m_{2*}.$$

(f) *Every future null geodesic cone $C^+(u)$ lying in the future of C_*^+ has negative expansion h in the exterior of a sphere $S_{\mathcal{A}}(u)$ lying in the interior of $S_2(u) = C_2^- \cap C^+(u)$ in $C^+(u)$.*

(g) *Every future null geodesic cone $C^+(u)$ lying in the future of C_*^+ intersects a singular boundary $\mathcal{B}$ at a sphere of zero radius.*

(h) *We define the apparent horizon $\mathcal{A}$ to be the union of all spheres with vanishing h. Then $\mathcal{A}$ is achronal and may contain outgoing null hypersurface pieces but no incoming null hypersurface pieces. In each outgoing null hypersurface piece ψ vanishes. In the past of $\mathcal{A}$ $h > 0$ while in the future of $\mathcal{A}$ $h < 0$.*

(i) *The apparent horizon $\mathcal{A}$ intersects the singular boundary $\mathcal{B}$ at $\mathcal{B}_0$, the set of singular boundary points at the center of symmetry. There is an incoming null hypersurface $\mathcal{N}_1$ which intersects $\mathcal{A}$ at $\mathcal{B}_0$. Every incoming null hypersurface $C^-(v)$ lying in the exterior of $\mathcal{N}_1$ intersects $\mathcal{A}$ at a sphere of positive radius and the singular boundary $\mathcal{B}$ at a sphere of zero radius.*

(j) *The singular boundary $\mathcal{A}$ is a future boundary; that is, there are no points whose past intersects $\mathcal{B}$. Furthermore, $\mathcal{B} \setminus \mathcal{B}_0$ is strictly spacelike.*

(k) *Let S be any sphere in $\mathcal{A}$. Then the mass contained within any sphere lying in the future of S is not less than the mass contained within S.*

(l) *The curvature invariant $|\mathrm{Riem}|^2$ blows up at $\mathcal{B}$ at a rate which is not slower than that of r^{-6}.*

Proof:

(a) Consider the spheres $S_2(u) = C_2^- \cap C^+(u)$. We shall show that if

$$\eta_0 \geqq E(\delta_0),$$

then $h_2(u)$ must become zero at or before the value u_1 of u at which

$$r_2(u_1) = \frac{3\delta_0}{(1 + \delta_0)} r_{2,0}.$$

Suppose on the contrary that $h_2(u) > 0$ for all $u \in [0, u_1]$. Consider the incoming null hypersurfaces $C^-(v)$ intersecting C_0^+ in spheres contained in the annular region

bounded by $S_{1,0}$ and $S_{2,0}$. Here v is the affine parameter value corresponding to $S_{0,v}$. We denote by v_1 and v_2 the affine parameter values corresponding to $S_{1,0}$ and $S_{2,0}$ respectively; we have $C^-(v_1) = C_1^-$, $C^-(v_2) = C_2^-$. Let

$$S(u, v) = C^+(u) \cap C^-(v)$$

and let

$$r(u, v) = \text{ radius of } S(u, v),$$

$$m(u, v) = \text{ mass content of } S(u, v),$$

$$r_1(u) = r(u, v_1) \quad , \quad r_2(u) = r(u, v_2),$$

$$m_1(u) = m(u, v_1) \quad , \quad m_2(u) = m(u, v_2).$$

[Note that here r and m are considered to be functions of (u, v), not of (u, s).] Then, by equation (4.4b) we have

$$\frac{\partial m}{\partial u} = \frac{1}{2} \underline{D} m = -\pi r^2 h \underline{\psi}^2$$

and

$$\frac{d}{du}(m_2 - m_1) = -\pi(h_2 r_2^2 \underline{\psi}_2^2 - h_1 r_1^2 \underline{\psi}_1^2).$$

Let

$$\eta = \frac{2(m_2 - m_1)}{r_2}.$$

Then, since

$$\frac{dr_2}{du} = \frac{1}{2} \underline{h}_2, \tag{5.1}$$

we obtain

$$\frac{d\eta}{du} = -\frac{2\pi}{r_2}(h_2 r_2^2 \underline{\psi}_2^2 - h_1 r_1^2 \underline{\psi}_1^2) - \frac{\underline{h}_2}{2r_2} \eta. \tag{5.2}$$

Let

$$\theta = r_2 \underline{\psi}_2 - r_1 \underline{\psi}_1.$$

Then we can express

$$h_2 r_2^2 \underline{\psi}_2^2 - h_1 r_1^2 \underline{\psi}_1^2 = h_2 \left[\theta^2 + 2\theta r_1 \underline{\psi}_1 - \left(\frac{h_1}{h_2} - 1 \right) r_1^2 \underline{\psi}_1^2 \right].$$

Now our hypothesis on h_2 allows us to apply Lemmas 2 and 3 in the interior of $S_2(u)$ in $C^+(u)$ for each $u \in [0, u_1)$. First, by virtue of Lemma 2 we obtain

$$\begin{aligned} & h_2 r_2^2 \underline{\psi}_2^2 - h_1 r_1^2 \underline{\psi}_1^2 \\ & \qquad \leqq h_2 [\theta^2 + 2\theta r_1 \underline{\psi}_1 - (e^\eta - 1) r_1^2 \underline{\psi}_1^2] \\ & \qquad \leqq h_2 \left(1 + \frac{1}{e^\eta - 1} \right) \theta^2. \end{aligned}$$

Then Lemma 3 yields the estimate

$$h_2 r_2^2 \underline{\psi}_2^2 - h_1 r_1^2 \underline{\psi}_1^2 \leqq - \frac{\underline{h}_2}{4\pi} \left(\frac{r_2}{r_1} - 1 \right) \eta \left(1 + \frac{1}{e^\eta - 1} \right). \tag{5.3}$$

Substituting this estimate in (5.2) we obtain

$$\frac{d\eta}{du} \geqq - \frac{\underline{h}_2}{2r_2} \eta \left[1 - \left(\frac{r_2}{r_1} - 1 \right) \left(1 + \frac{1}{e^\eta - 1} \right) \right]. \tag{5.4}$$

We now define

$$x = \frac{r_2(u)}{r_2(0)}.$$

Our hypothesis allows us to apply Property 4.2 to the null hypersurfaces C_1^-, C_2^-, C_0^+, and $C^+(u)$ for any $u \in [0, u_1]$ to conclude that

$$\frac{r_2(u) - r_1(u)}{r_2(0) - r_1(0)} \leqq 1.$$

It follows that

$$\delta(u) := \frac{r_2(u)}{r_1(u)} - 1 = \frac{r_2(u) - r_1(u)}{r_2(u) - (r_2(u) - r_1(u))} \tag{5.5}$$

$$\leqq \frac{r_2(0) - r_1(0)}{r_2(u) - (r_2(0) - r_1(0))} = \frac{\delta_0}{x(1 + \delta_0) - \delta_0}. \tag{5.6}$$

Substituting this in (5.4) and taking into account (5.1) we arrive at the following ordinary differential inequality:

$$-x\frac{d\eta}{dx} \geqq \eta\left[1 - \frac{\delta_0}{x(1+\delta_0)-\delta_0}\left(1 + \frac{1}{e^{\eta}-1}\right)\right]. \tag{5.7a}$$

Since for $\eta \geqq 0 : e^{\eta} - 1 \geqq \eta$ we can simplify this to

$$-x\frac{d\eta}{dx} \geqq g\eta - f \tag{5.7b}$$

where

$$g(x) = 1 - \frac{\delta_0}{x(1+\delta_0)-\delta_0},$$

$$f(x) = \frac{\delta_0}{x(1+\delta_0)-\delta_0}.$$

Integrating (5.7b) on $[x, 1]$ we obtain

$$\eta(x) \geqq e^{G(x)}(\eta_0 - F(x)) \tag{5.8}$$

where

$$G(x) = \int_x^1 g(x')\frac{dx'}{x'},$$

$$F(x) = \int_x^1 e^{-G(x')}f(x')\frac{dx'}{x'}.$$

We have

$$G(x) = \log\left[\frac{x(1+\delta_0)-\delta_0}{x^2}\right],$$

$$F(x) = \frac{\delta_0}{(1+\delta_0)^2}\left\{\log\left[\frac{1}{x(1+\delta_0)-\delta_0}\right] + \delta_0\left[\frac{1}{x(1+\delta_0)-\delta_0} - 1\right]\right\}.$$

The function

$$F + e^{-G}$$

achieves its minimum on $(0, 1]$ at

$$x_1 = \frac{3\delta_0}{(1 + \delta_0)}$$

which corresponds to the value u_1 of u. The minimum value is

$$F(x_1) + e^{-G(x_1)} = E(\delta_0).$$

Now, the hypothesis $h_2(u_1) > 0$ implies $(2m_2(u_1))/(r_2(u_1)) < 1$, hence, a fortiori, $\eta(x_1) < 1$. But then (5.8) yields

$$\eta_0 < F(x_1) + e^{-G(x_1)} = E(\delta_0),$$

contradicting the assumption of the theorem. It follows that there is a $u_* \in [0, u_1]$ where h_2 becomes zero. We denote $C^+(u_*) = C^+_*$.

(b) We have

$$r_{2*} \geqq r(u_1) = \frac{3\delta_0}{(1 + \delta_0)} r_{2,0},$$

and the fact that $h_{2*} = 0$ implies by (4.3)

$$m_{2*} = \frac{r_{2*}}{2}.$$

(c) Let $S_{1*} = C_1^- \cap C^+_*$ and let r_{1*} be the radius of S_{1*}. Then, by Proposition 4.2,

$$r_{1*} = r_{2*} - (r_{2*} - r_{1*}) \geqq r_{2*} - (r_{2,0} - r_{1,0})$$

$$\geqq 3\delta_0 r_{1,0} - \delta_0 r_{1,0} = 2\delta_0 r_{1,0}.$$

(d) By (5.1) and Lemma 1,

$$-\frac{dr_2}{du} \geqq \frac{1}{2}.$$

Integrating on $[0, u_*]$ we obtain

$$u_* \leqq 2\left(1 - \frac{3\delta_0}{1 + \delta_0}\right) r_{2,0} = 2(1 - 2\delta_0) r_{1,0}.$$

(e) Clearly u_* is an upper bound for the set U. Hence

$$u_{\mathscr{S}} \leqq u_*.$$

By (4.4a) and Property 4.3,

$$M_1 \geqq m_2(u_{\mathscr{I}}),$$

while by (4.4b) and the fact that $h_2 \geqq 0$ on $[0, u_*]$,

$$m_2(u_{\mathscr{I}}) \geqq m_{2*}.$$

Hence

$$M_1 \geqq m_{2*}.$$

(f), (g) By equations (4.3), (4.1c), and (4.4b),

$$\underline{D}(h\underline{h}) = -\frac{2m}{r^2}\underline{h} - 4\pi h\underline{\psi}^2. \tag{5.9}$$

Consider in general an incoming null hypersurface C^- and the spheres $S(u) = C^- \cap C^+(u)$. Let $r(u)$ be the radius of $S(u)$ and let $u_{\mathscr{B}}$ be the value of u at which $r(u_{\mathscr{B}}) = 0$. Suppose that the sphere $S(u_{\mathscr{A}})$ has $h(u_{\mathscr{A}}) = 0$. Then for all $u \in (u_{\mathscr{A}}, u_{\mathscr{B}})$ the sphere $S(u)$ has $h(u) < 0$. For, by equation (5.9), since

$$\left(\frac{2m}{r}\right)\Big|_{\mathscr{A}} = 1,$$

we have

$$\underline{D}(h\underline{h})|_{\mathscr{A}} = -\left(\frac{\underline{h}}{r}\right)\Big|_{\mathscr{A}} > 0.$$

Hence there is an $\epsilon > 0$ such that

$$h(u) < 0 \quad : \quad \forall u \in [u_{\mathscr{A}}, u_{\mathscr{A}} + \epsilon].$$

Let

$$u_2 = \sup\{u \in (u_{\mathscr{A}}, u_{\mathscr{B}}) \quad : \quad \forall u' \in (u_{\mathscr{A}}, u] \quad : \quad h(u') < 0\}.$$

By equation (4.4a) in $[u_{\mathscr{A}}, u_2]$ we have $\underline{D}m \geqq 0$. Now, either $u_2 = u_{\mathscr{B}}$ or $h(u_2) = 0$. From equation (5.9), for all $u \in [u_{\mathscr{A}}, u_2]$ we have

$$\underline{D}(h\underline{h}) \geqq -\frac{2m}{r^2}\underline{h} \geqq -\frac{r_{\mathscr{A}}}{r^2}\underline{h},$$

as $m \geqq m_{\mathscr{A}} = \frac{1}{2}r_{\mathscr{A}}$. Thus, integrating along C^- (noting that $\underline{D}r = \underline{h}$), we obtain

$$h\underline{h} \geqq \frac{r_{\mathscr{A}}}{r} - 1$$

or, by (4.2c),

$$h \leqq -\frac{(r_{\mathcal{A}} - r)}{s}. \tag{5.10}$$

This eliminates the alternative $h(u_2) = 0$. We conclude that for all $u \in (u_{\mathcal{A}}, u_{\mathcal{B}})$ inequality (5.10) holds along C^-. Now according to equation (4.1d) h is decreasing along each future cone. It follows that for each $u \in (u_{\mathcal{A}}, u_{\mathcal{B}})$ the future null geodesic cone $C^+(u)$ has negative expansion in the exterior of a sphere which lies in the interior of $S(u)$. In particular, $h < 0$ in the future of the sphere $S(u_{\mathcal{A}})$. Also, for each $u \in (u_{\mathcal{A}}, u_{\mathcal{B}})$ in the exterior of $S(u)$ in $C^+(u)$ it holds:

$$r(u, s) \leqq r(u) + h(u)(s - s(u))$$

where $s(u)$ is the affine parameter value corresponding to $S(u)$. We conclude that r becomes zero within affine distance

$$\Delta s = -\frac{r(u)}{h(u)} \tag{5.11}$$

from $S(u)$.

The above apply in particular to the incoming null hypersurface C_2^- for which $u_{\mathcal{A}} = u_*$.

(h) The argument in the previous paragraph shows that if S is any sphere in $\mathcal{A}$, then $h < 0$ in the interior of the future of S as well as on the incoming part of the boundary of the future of S. Thus $\mathcal{A}$ does not intersect the future of S except possibly the outgoing part C^+ of the boundary of the future of S. It follows that $\mathcal{A}$ is achronal. Suppose that $\mathcal{A}$ intersects C^+ at S'. Then h vanishes in the annular region in C^+ bounded by S and S'. Therefore by equation (4.1d) ψ vanishes in this region.

(i) $\mathcal{A}$ cannot intersect $\mathcal{B}$ except at $\mathcal{B}_0$. For, suppose that $\mathcal{A}$ intersects $\mathcal{B}$ at a sphere S_0 where the future null geodesic cone C^+ intersects $\mathcal{B}$. Then $h > 0$ on C^+ while the radius of S_0 vanishes: a contradiction. On the other hand, $\mathcal{A}$ must intersect $\mathcal{B}$. For, if not, the central point of $\mathcal{A}$ has a future; but this is impossible since r vanishes at the vertex while $h < 0$ in the future of $\mathcal{A}$. Now, let $C^-(v)$ be an incoming null hypersurface which intersects $\mathcal{A}$. Then for each $v' \geqq v$ $C(v')$ also intersects $\mathcal{A}$. We define:

$$v_{\mathcal{N}_1} = \inf\{v : C^-(v) \quad \text{intersects} \quad \mathcal{A}\}.$$

[Note: $v_{\mathcal{N}_1} > 0$ since $\mathcal{A}$ does not intersect C_0^+.] Then $C^-(v_{\mathcal{N}_1}) = \mathcal{N}_1$ intersects $\mathcal{A}$ at $\mathcal{B}_0$ and every incoming null hypersurface $C^-(v)$ in the exterior of $\mathcal{N}_1$ intersects $\mathcal{A}$ at a sphere $S_{\mathcal{A}}(v)$ of positive radius $r_{\mathcal{A}}(v)$ after which within

$$\Delta u = 2r_{\mathcal{A}},$$

$C^-(v)$ intersects $\mathcal{B}$ at a sphere of zero radius.

Remark. We can also define

$$v_{\mathcal{N}_0} = \sup\{v : C^-(v) \quad \text{intersects} \quad \Gamma\}.$$

Then $C^-(v_{\mathcal{N}_0}) = \mathcal{N}_0$ intersects $\mathcal{B}_0$. Now $\mathcal{N}_1$ is exterior to $\mathcal{N}_0$ but does not in general coincide with $\mathcal{N}_0$. The incoming null hypersurfaces $C^-(v)$ with $v \in [v_{\mathcal{N}_0}, v_{\mathcal{N}_1}]$ are trapped between $\mathcal{N}_0$ and $\mathcal{N}_1$, hence they also intersect $\mathcal{B}_0$.

(j) Suppose that there is a sphere S whose past intersects $\mathcal{B}$. Then either C^+ or C^-, respectively the outgoing and incoming null hypersurfaces through S drawn toward the past, intersects $\mathcal{B}$. The first case is impossible since r vanishes at the intersection of C^+ with $\mathcal{B}$ and $r(S) > 0$, while $Dr = h < 0$ as C^+ lies in the future of $\mathcal{A}$. The second case is also impossible as $\underline{D}r = \underline{h} < 0$.

Now let an incoming null hypersurface $C^-(v)$ intersect $\mathcal{B}$ at the value $u_{\mathcal{B}}(v)$ of u. Then for $u \leqq u_{\mathcal{B}}(v)$ we have, along $C^-(v)$,

$$r(u) \geqq \tfrac{1}{2}(u_{\mathcal{B}} - u).$$

This shows that $\mathcal{B}$ does not contain incoming null pieces. To show that $\mathcal{B} \setminus \mathcal{B}_0$ does not contain outgoing null pieces, consider two incoming null hypersurfaces $C^-(v_1)$ and $C^-(v_2)$ intersecting $\mathcal{B} \setminus \mathcal{B}_0$ with $C^-(v_2)$ in the exterior of $C^-(v_1)$. Let $C^-(v_2)$ intersect $\mathcal{B} \setminus \mathcal{B}_0$ at the value $u_{\mathcal{B}}(v_2) = u_2$ of u. Consider a future null geodesic cone $C^+(u_1)$ in the past of $C^+(u_2)$. Then if u_1 is sufficiently close to u_2 and v_1 is sufficiently close to v_2, the sphere $S(u_1, v_1)$ lies in the future of $\mathcal{A}$. Hence $h < 0$ in the spacetime region bounded by $C^+(u_1)$, $C^+(u_2)$, $C^-(v_1)$, $C^-(v_2)$. Therefore

$$f = \frac{1}{h}\frac{m}{r^2} < 0$$

in this region. Equation (4.7) then implies that

$$\frac{\partial r}{\partial v}(u_2, v) < \frac{\partial r}{\partial v}(u_1, v)$$

for all incoming null hypersurfaces $C^-(v)$ between $C^-(v_1)$ and $C^-(v_2)$. Integrating on $[v_1, v_2]$ we then obtain

$$r(u_2, v_1) > r(u_1, v_1) - r(u_1, v_2).$$

This shows in particular that $C^-(v_1)$ intersects $\mathcal{B} \setminus \mathcal{B}_0$ at a value of u which is strictly greater than u_2.

(k) Let $S_{\mathcal{A}}$ be any sphere on $\mathcal{A}$ and let C^- and C^+ be the incoming and outgoing null hypersurfaces through $S_{\mathcal{A}}$ drawn toward the future. We have shown

above that $h < 0$ everywhere in the future of $S_{\mathcal{A}}$ except possibly on C^+ where $h \leqq 0$. It then follows from equations (4.4a), (4.4b) that the mass m of every sphere S in the future of $S_{\mathcal{A}}$ is not less than the mass of $S_{\mathcal{A}}$. On the other hand,

$$m_{\mathcal{A}} = \tfrac{1}{2} r_{\mathcal{A}} > 0.$$

(1) Consider the future of $S_{\mathcal{A}}$. By equation (3.7),

$$\frac{1}{4} R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta} \geqq 2\left(\frac{2m}{r^3}\right)^2.$$

We conclude that in the future of $S_{\mathcal{A}}$:

$$|\mathrm{Riem}|^2 \geqq 8\left(\frac{r_{\mathcal{A}}}{r^3}\right)^2.$$

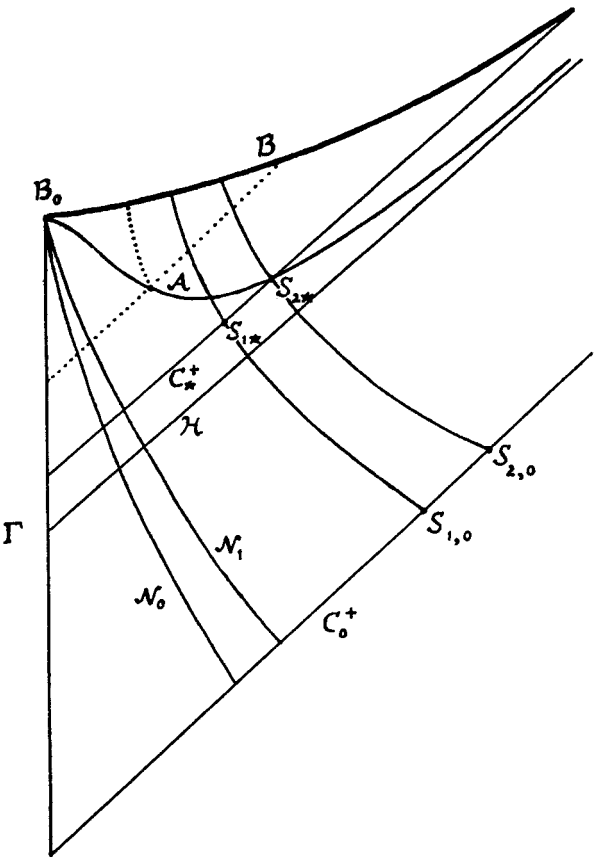


Figure 1.

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